

Generate tables of values from visually growing patterns or the rule of a function; describe and plot these relationships on the Cartesian plane

Learning intention

- generate tables of values from visually growing patterns or the rule of a function; describe and plot these relationships on the Cartesian plane.

Teach and model

- using diagrams and manipulatives to form linear growth patterns.
- using function machines to generate a table of ordered pairs using input and output values.
- plotting points from a table of values generated using simple linear functions and recognising patterns, such as the points lie on a straight line.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.